UC Berkeley Math 128A: Homework 1

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MATLAB usage classification: PEN (pen and paper only), CALC (use as calculator), CODE (run codes).

0. Please assign yourself a self grade on this homework (0, 1, or 2) based off the guidelines in the syllabus.

1. (1.1-1c, CALC) Show that the following equation has at least one solution in the given intervals:

$$2x\cos(2x) - (x-2)^2 = 0, \qquad [2,3] \text{ and } [3,4].$$

2. (1.1-5a, PEN) Find $\max_{0\le x\le 1} |f(x)|$ when $f(x) = (2 - e^x + 2x)/3$.

3. (1.1-16, CALC) Use the error term of a Taylor polynomial to estimate the error involved in using $\sin x \approx x$ to approximate $\sin 1^\circ$.

4. (1.1-19, CALC) Let $f(x) = e^x$ and $x_0 = 0$. Find the nth Taylor polynomial $P_n(x)$ for $f(x)$ about x_0. Find a value of n necessary for $P_n(x)$ to approximate $f(x)$ to within 10^{-6} on $[0, 0.5]$.

5. (1.1-22, PEN) Use the Intermediate Value Theorem and Rolle's theorem to show that the graph of $f(x) = x^3 + 2x + k$ crosses the x-axis exactly once, regardless of the value of the constant k.

6. (1.1-29b, PEN) Suppose that f is continuous on $[a,b]$, that x_1 and x_2 are in $[a,b]$, and that c_1 and c_2 are positive constants. Show that a number ξ exists between x_1 and x_2 with

$$f(\xi) = \frac{c_1 f(x_1) + c_2 f(x_2)}{c_1 + c_2}.$$

7. (1.2-1a/2d, CALC) Compute the absolute error and relative error in approximations of p by p^* :
1a. $p = \pi, p^* = 22/7$
2d. $p = 9!, p^* = \sqrt{18\pi}(9/e)^9$

8. (1.2-5c, CALC) Perform the following computation (i) exactly, (ii) using three-digit chopping arithmetic, and (iii) using three-digit rounding arithmetic. (iv) Compute the relative errors in parts (ii) and (iii).

$$\left(\frac{1}{3} - \frac{3}{11}\right) + \frac{3}{20}$$

9. (1.2-11, CALC) The first three nonzero terms of the Maclaurin series for the arctangent function are $x - (1/3)x^3 + (1/5)x^5$. Compute the absolute error and relative error in the following approximations of π using the polynomial in place of the arctangent:

a.

$$4\left[\arctan\left(\frac{1}{2}\right) + \arctan\left(\frac{1}{3}\right)\right]$$

b.

$$16\arctan\left(\frac{1}{5}\right) - 4\arctan\left(\frac{1}{239}\right)$$

10. (1.2-14, CALC) Let

$$f(x) = \frac{e^x - e^{-x}}{x}$$

a. Find $\lim_{x\to 0}\left(e^x - e^{-x}\right)/x$.
b. Use three-digit rounding arithmetic to evaluate $f(0.1)$.
c. Replace each exponential function with its third Maclaurin polynomial, simplify, and repeat part (b).
d. The actual value is $f(0.1) = 2.003335000$. Find the relative error for the values obtained in parts (b) and (c).

11. (1.2-19/20cd, CALC, use Wolfram Alpha for exact decimal numbers) Use the 64-bit long real format to find the decimal equivalent of the following floating-point machine numbers. Find the next largest and smallest machine numbers in decimal form for these numbers.

c. 0 01111111111 0101001100000000000000000000000000000000000000000000

d. 0 01111111111 0101001100000000000000000000000000000000000000000001